Lemma 1. • *The distribution function of the k-th uniform record value* $U_{r,\tilde{\gamma}_r} = 1 - \prod_{j=1}^{r} U_j^{1/k}$ *is given by:* $F_{U_{r,\tilde{\gamma}_r}}(u) = F_{\mathrm{Gamma}(r,1)}(-k\ln(1-u))$.

- *The density* $f_{U_{r,\tilde{\gamma}_r}}$ *is bounded if and only if* $k \geq 2 \quad$ *or* $\quad (k = 1,\ r = 1)$.

Proof of Lemma 1. • The CDF can be expressed as:

$$\begin{aligned} F_{U_{r,\tilde{\gamma}_r}}(u) &= P\left(1 - \prod_{j=1}^{r} U_j^{1/k} \leq u\right) \\ &= P\left(\prod_{j=1}^{r} U_j \geq (1-u)^k\right) \\ &= P\left(-\sum_{j=1}^{r} \ln U_j \leq -k\ln(1-u)\right). \end{aligned}$$

Since $U_1, \ldots, U_r$ are i.i.d. Uniform$(0,1)$ random variables, the transformed random variables $-\ln U_1, \ldots, -\ln U_r$ are independent and each follows an exponential distribution with rate 1. Consequently, the sum is of Gamma$(r,1)$ distribution: $-\sum_{j=1}^{r} \ln U_j \sim \mathrm{Gamma}(r,1)$.

- Differentiating the CDF with respect to u gives the PDF.

$$f_{U_{r,\tilde{\gamma}_r}}(u) = \begin{cases} \dfrac{k^r}{(r-1)!}(1-u)^{k-1}\big[-\ln(1-u)\big]^{r-1}, & 0 < u < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Since $f_{U_{r,\tilde{\gamma}_r}}$ is continuous on $(0,1)$, its boundedness can be determined by examining its behavior at the endpoints. As $r \geq 1$, the density satisfies

$$\lim_{u\to 0^+} f_{U_{r,\tilde{\gamma}_r}}(u) = \begin{cases} k, & r = 1, \\ 0, & r > 1, \end{cases}$$

and is therefore bounded in a neighborhood of $u = 0$. Consequently, it remains to investigate the behavior of the density as $u \to 1^-$ to determine whether it is bounded on $(0,1)$.

Next, setting $t = -\ln(1-u)$, we can rewrite the density as

$$f_{U_{r,\tilde{\gamma}_r}}(u) = \frac{k^r}{(r-1)!} e^{-(k-1)t} t^{r-1}, \qquad t \to \infty.$$

If $k > 1$, the exponential decay dominates the polynomial term, and therefore

$$\lim_{u\to 1^-} f_{U_{r,\tilde{\gamma}_r}}(u) = 0.$$

Since the density is continuous on $(0,1)$ and has finite limits at both endpoints, it is bounded.

If $k = 1$, the density reduces to

$$f_{U_{r,\tilde{\gamma}_r}}(u) = \frac{[-\ln(1-u)]^{r-1}}{(r-1)!}.$$

Hence,

$$\lim_{u \to 1^-} f_{U_{r,\tilde{\gamma}_r}}(u) = \begin{cases} 1, & r = 1, \\ +\infty, & r > 1. \end{cases}$$

□

Proposition 2. *Let $\pi^G_{r,\tilde{\gamma}_r} = G(\mathbb{E}(G^{-1}(U_{r,\tilde{\gamma}_r})))$. Then, for every r and $\tilde{\gamma}_r$ such that $\mathbb{E}(G^{-1}(U_{r,\tilde{\gamma}_r}))$ is defined, we have the following:*

1. *if $F \in \mathcal{F}^{cx}_G$, then $P(X \leq \mu_{r,\tilde{\gamma}_r}) \leq \pi^G_{r,\tilde{\gamma}_r}$;*
2. *if $F \in \mathcal{F}^{cv}_G$, then $P(X \leq \mu_{r,\tilde{\gamma}_r}) \geq \pi^G_{r,\tilde{\gamma}_r}$.*